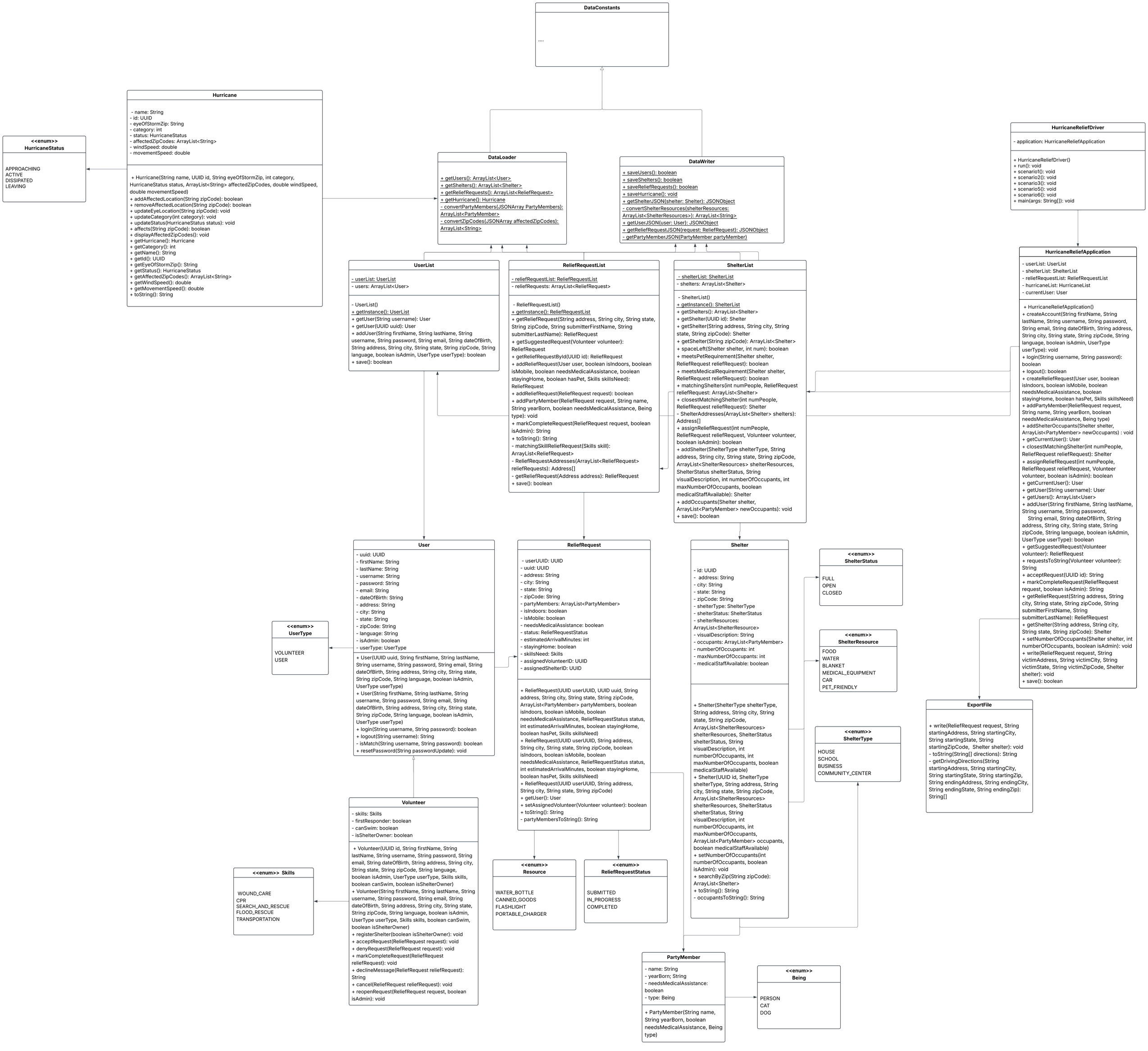

DataConstants
....
Hurricane
- name: String
- id: UUID
- eyeOfStormZip: String
- category: int
- status: HurricaneStatus
- affectedZipCodes: ArrayList<String>
- windSpeed: double
- movementSpeed: double
+ Hurricane(String name, UUID id, String eyeOfStormZip, int category, HurricaneStatus status, ArrayList<String> affectedZipCodes, double windSpeed, double movementSpeed)
+ addAffectedLocation(String zipCode): boolean
+ removeAffectedLocation(String zipCode): boolean
+ updateEyeLocation(String zipCode): void
+ updateCategory(int category): void
+ updateStatus(HurricaneStatus status): void
+ affects(String zipCode): boolean
+ displayAffectedZipCodes(): void
+ getHurricane(): Hurricane
+ getCategory(): int
+ getName(): String
+ getId(): UUID
+ getEyeOfStormZip(): String
+ getStatus(): HurricaneStatus
+ getAffectedZipCodes(): ArrayList<String>
+ getWindSpeed(): double
+ getMovementSpeed(): double
+ toString(): String
<<enum>>
HurricaneStatus
APPROACHING
ACTIVE
DISSIPATED
LEAVING
DataLoader
+ getUsers(): ArrayList<User>
+ getShelters(): ArrayList<Shelter>
+ getReliefRequests(): ArrayList<ReliefRequest>
+ getHurricane(): Hurricane
- convertPartyMembers(JSONArray PartyMembers): ArrayList<PartyMember>
- convertZipCodes(JSONArray affectedZipCodes): ArrayList<String>
DataWriter
+ saveUsers(): boolean
+ saveShelters(): boolean
+ saveReliefRequests(): boolean
+ saveHurricane(): void
+ getShelterJSON(shelter: Shelter): JSONObject
- convertShelterResources(shelterResources: ArrayList<ShelterResources>): ArrayList<String>
+ getUserJSON(user: User): JSONObject
+ getReliefRequestJSON(request: ReliefRequest): JSONObject
- getPartyMemberJSON(PartyMember partyMember)
HurricaneReliefDriver
- application: HurricaneReliefApplication
+ HurricaneReliefDriver()
+ run(): void
+ scenario1(): void
+ scenario2(): void
+ scenario3(): void
+ scenario5(): void
+ scenario6(): void
+ main(args: String[]): void
UserList
- userList: UserList
- users: ArrayList<User>
- UserList()
+ getInstance(): UserList
+ getUser(String username): User
+ getUser(UUID uuid): User
+ addUser(String firstName, String lastName, String username, String password, String email, String dateOfBirth, String address, String city, String state, String zipCode, String language, boolean isAdmin, UserType userType): boolean
+ save(): boolean
ReliefRequestList
- reliefRequestList: ReliefRequestList
- reliefRequests: ArrayList<ReliefRequest>
- ReliefRequestList()
+ getInstance(): ReliefRequestList
+ getReliefRequest(String address, String city, String state, String zipCode, String submitterFirstName, String submitterLastName): ReliefRequest
+ getSuggestedRequest(Volunteer volunteer): ReliefRequest
+ getReliefRequestById(UUID id): ReliefRequest
+ addReliefRequest(User user, boolean isIndoors, boolean isMobile, boolean needsMedicalAssistance, boolean stayingHome, boolean hasPet, Skills skillsNeed): ReliefRequest
+ addReliefRequest(ReliefRequest request): boolean
+ addPartyMember(ReliefRequest request, String name, String yearBorn, boolean needsMedicalAssistance, Being type): void
+ markCompleteRequest(ReliefRequest request, boolean isAdmin): String
+ toString(): String
- matchingSkillReliefRequest(Skills skill): ArrayList<ReliefRequest>
- ReliefRequestAddresses(ArrayList<ReliefRequest> reliefRequests): Address[]
- getReliefRequest(Address address): ReliefRequest
+ save(): boolean
ShelterList
- shelterList: ShelterList
- shelters: ArrayList<Shelter>
- ShelterList()
+ getInstance(): ShelterList
+ getShelters(): ArrayList<Shelter>
+ getShelter(UUID id): Shelter
+ getShelter(String address, String city, String state, String zipCode): Shelter
+ getShelter(String zipCode): ArrayList<Shelter>
+ spaceLeft(Shelter shelter, int num): boolean
+ meetsPetRequirement(Shelter shelter, ReliefRequest reliefRequest): boolean
+ meetsMedicalRequirement(Shelter shelter, ReliefRequest reliefRequest): boolean
+ matchingShelters(int numPeople, ReliefRequest reliefRequest: ArrayList<Shelter>
+ closestMatchingShelter(int numPeople, ReliefRequest reliefRequest): Shelter
- ShelterAddresses(ArrayList<Shelter> shelters): Address[]
+ assignReliefRequest(int numPeople, ReliefRequest reliefRequest, Volunteer volunteer, boolean isAdmin): boolean
+ addShelter(ShelterType shelterType, String address, String city, String state, String zipCode, ArrayList<ShelterResources> shelterResources, ShelterStatus shelterStatus, String visualDescription, int numberOfOccupants, int maxNumberOfOccupants, boolean medicalStaffAvailable): Shelter
+ addOccupants(Shelter shelter, ArrayList<PartyMember> newOccupants): void
+ save(): boolean
HurricaneReliefApplication
- userList: UserList
- shelterList: ShelterList
- reliefRequestList: ReliefRequestList
- hurricaneList: HurricaneList
- currentUser: User
+ HurricaneReliefApplication()
+ createAccount(String firstName, String lastName, String username, String password, String email, String dateOfBirth, String address, String city, String state, String zipCode, String language, boolean isAdmin, UserType userType): void
+ login(String username, String password): boolean
+ logout(): boolean
+ createReliefRequest(User user, boolean isIndoors, boolean isMobile, boolean needsMedicalAssistance, boolean stayingHome, boolean hasPet, Skills skillsNeed)
+ addPartyMember(ReliefRequest request, String name, String yearBorn, boolean needsMedicalAssistance, Being type)
+ addShelterOccupants(Shelter shelter, ArrayList<PartyMember> newOccupants) : void
+ getCurrentUser(): User
+ closestMatchingShelter(int numPeople, ReliefRequest reliefRequest): Shelter
+ assignReliefRequest(int numPeople, ReliefRequest reliefRequest, Volunteer volunteer, boolean isAdmin): boolean
+ getCurrentUser(): User
+ getUser(String username): User
+ getUsers(): ArrayList<User>
+ addUser(String firstName, String lastName, String username, String password, String email, String dateOfBirth, String address, String city, String state, String zipCode, String language, boolean isAdmin, UserType userType): boolean
+ getSuggestedRequest(Volunteer volunteer): ReliefRequest
+ requeststoString(Volunteer volunteer): String
+ acceptRequest(UUID id): String
+ markCompleteRequest(ReliefRequest request, boolean isAdmin): String
+ getReliefRequest(String address, String city, String state, String zipCode, String submitterFirstName, String submitterLastName): ReliefRequest
+ getShelter(String address, String city, String state, String zipCode): Shelter
+ setNumberOfOccupants(Shelter shelter, int numberOfOccupants, boolean isAdmin): void
+ write(ReliefRequest request, String victimAddress, String victimCity, String victimState, String victimZipCode, Shelter shelter): void
+ save(): boolean
User
- uuid: UUID
- firstName: String
- lastName: String
- username: String
- password: String
- email: String
- dateOfBirth: String
- address: String
- city: String
- state: String
- zipCode: String
- language: String
- isAdmin: boolean
- userType: UserType
+ User(UUID uuid, String firstName, String lastName, String username, String password, String email, String dateOfBirth, String address, String city, String state, String zipCode, String language, boolean isAdmin, UserType userType)
+ User(String firstName, String lastName, String username, String password, String email, String dateOfBirth, String address, String city, String state, String zipCode, String language, boolean isAdmin, UserType userType)
+ login(String username, String password): boolean
+ logout(String username): String
- isMatch(String username, String password): boolean
+ resetPassword(String passwordUpdate): void
<<enum>>
UserType
VOLUNTEER
USER
ReliefRequest
- userUUID: UUID
- uuid: UUID
- address: String
- city: String
- state: String
- zipCode: String
- partyMembers: ArrayList<PartyMember>
- isIndoors: boolean
- isMobile: boolean
- needsMedicalAssistance: boolean
- status: ReliefRequestStatus
- estimatedArrivalMinutes: int
- stayingHome: boolean
- skillsNeed: Skills
- assignedVolunteerID: UUID
- assignedShelterID: UUID
+ ReliefRequest(UUID userUUID, UUID uuid, String address, String city, String state, String zipCode, ArrayList<PartyMember> partyMembers, boolean isIndoors, boolean isMobile, boolean needsMedicalAssistance, ReliefRequestStatus status, int estimatedArrivalMinutes, boolean stayingHome, boolean hasPet, Skills skillsNeed)
+ ReliefRequest(UUID userUUID, String address, String city, String state, String zipCode, boolean isIndoors, boolean isMobile, boolean needsMedicalAssistance, ReliefRequestStatus status, int estimatedArrivalMinutes, boolean stayingHome, boolean hasPet, Skills skillsNeed)
+ ReliefRequest(UUID userUUID, String address, String city, String state, String zipCode)
+ getUser(): User
+ setAssignedVolunteer(Volunteer volunteer): boolean
+ toString(): String
- partyMembersToString(): String
Shelter
- id: UUID
- address: String
- city: String
- state: String
- zipCode: String
- shelterType: ShelterType
- shelterStatus: ShelterStatus
- shelterResources: ArrayList<ShelterResource>
- visualDescription: String
- occupants: ArrayList<PartyMember>
- numberOfOccupants: int
- maxNumberOfOccupants: int
- medicalStaffAvailable: boolean
+ Shelter(ShelterType shelterType, String address, String city, String state, String zipCode, ArrayList<ShelterResources> shelterResources, ShelterStatus shelterStatus, String visualDescription, int numberOfOccupants, int maxNumberOfOccupants, boolean medicalStaffAvailable)
+ Shelter(UUID id, ShelterType shelterType, String address, String city, String state, String zipCode, ArrayList<ShelterResources> shelterResources, ShelterStatus shelterStatus, String visualDescription, int numberOfOccupants, int maxNumberOfOccupants, ArrayList<PartyMember> occupants, boolean medicalStaffAvailable)
+ setNumberOfOccupants(int numberOfOccupants, boolean isAdmin): void
+ searchByZip(String zipCode): ArrayList<Shelter>
+ toString(): String
- occupantsToString(): String
<<enum>>
ShelterStatus
FULL
OPEN
CLOSED
<<enum>>
ShelterResource
FOOD
WATER
BLANKET
MEDICAL_EQUIPMENT
CAR
PET_FRIENDLY
<<enum>>
ShelterType
HOUSE
SCHOOL
BUSINESS
COMMUNITY_CENTER
ExportFile
+ write(ReliefRequest request, String startingAddress, String startingCity, String startingState, String startingZipCode, Shelter shelter): void
- toString(String[] directions): String
- getDrivingDirections(String startingAddress, String startingCity, String startingState, String startingZip, String endingAddress, String endingCity, String endingState, String endingZip): String[]
Volunteer
- skills: Skills
- firstResponder: boolean
- canSwim: boolean
- isShelterOwner: boolean
+ Volunteer(UUID id, String firstName, String lastName, String username, String password, String email, String dateOfBirth, String address, String city, String state, String zipCode, String language, boolean isAdmin, UserType userType, Skills skills, boolean canSwim, boolean isShelterOwner)
+ Volunteer(String firstName, String lastName, String username, String password, String email, String dateOfBirth, String address, String city, String state, String zipCode, String language, boolean isAdmin, UserType userType, Skills skills, boolean canSwim, boolean isShelterOwner)
+ registerShelter(boolean isShelterOwner): void
+ acceptRequest(ReliefRequest request): void
+ denyRequest(ReliefRequest request): void
+ markCompleteRequest(ReliefRequest reliefRequest): void
+ declineMessage(ReliefRequest reliefRequest): String
+ cancel(ReliefRequest reliefRequest): void
+ reopenRequest(ReliefRequest request, boolean isAdmin): void
<<enum>> Skills
WOUND_CARE
CPR
SEARCH_AND_RESCUE
FLOOD_RESCUE
TRANSPORTATION
<<enum>>
Resource
WATER_BOTTLE
CANNED_GOODS
FLASHLIGHT
PORTABLE_CHARGER
<<enum>>
ReliefRequestStatus
SUBMITTED
IN_PROGRESS
COMPLETED
PartyMember
- name: String
- yearBorn; String
- needsMedicalAssistance: boolean
- type: Being
+ PartyMember(String name, String yearBorn, boolean needsMedicalAssistance, Being type)
<<enum>>
Being
PERSON
CAT
DOG